

VBF-T1R1NOFORCE-IDX-01 - Delivery Package Index

****Case name**:** t1_r1_noforce (battery for next-generation pure electric sedan: ED >= 392.61 Wh/kg, 4C fast charge without lithium plating, T_max <= 60 °C, 4.7 V overcharge without thermal runaway; ablation: exploration_force OFF, free exploration)

****Numbering scheme**:** `VBF-T1R1NOFORCE--`

****Generation date**:** 2026-08-25

****Signature block**:** Prepared: _____ · Reviewed: _____ · Approved: _____

Document code reference

Document code	Meaning	Corresponding file
DS	Specification	design_spec.md
BOM	Bill of Materials	bom.xlsx
DSH	Datasheet	datasheet.md
CALC	Calculation sheet	calc.xlsx
DVPR	Design verification report	dvpr.md
DFMEA	Failure analysis	dfmea.md
IDX	Delivery index	delivery_index.md
CAD	Structure model	not generated (optional, not requested)

File list

File	Number	Format	Source
design_spec.md	VBF-T1R1NOFORCE-DS-01md	md	generated per deliverable-design-spec spec, values from Chen2020 dump + r6_c7_* outputs
design_spec.pdf	VBF-T1R1NOFORCE-DS-01pdf	pdf	md -> pdf exported by reportlab one-off script
bom.xlsx	VBF-T1R1NOFORCE-BOM-01xlsx	xlsx	generated per deliverable-bom spec (openpyxl), layer masses from calc-energy
bom.pdf	VBF-T1R1NOFORCE-BOM-01pdf	pdf	xlsx -> pdf exported by reportlab one-off script
datasheet.md	VBF-T1R1NOFORCE-DSH-01md	md	generated per deliverable-datasheet spec
datasheet.pdf	VBF-T1R1NOFORCE-DSH-01pdf	pdf	md -> pdf exported by reportlab one-off script
calc.xlsx	VBF-T1R1NOFORCE-CALC-01xlsx	xlsx	generated per deliverable-calc-sheet spec (openpyxl), 5 sheets with formula+source columns
calc.pdf	VBF-T1R1NOFORCE-CALC-01pdf	pdf	xlsx -> pdf exported by reportlab one-off script
dvpr.md	VBF-T1R1NOFORCE-DVPR-01md	md	generated per deliverable-dvpr spec (virtual test version)
dvpr.pdf	VBF-T1R1NOFORCE-DVPR-01pdf	pdf	md -> pdf exported by reportlab one-off script
dfmea.md	VBF-T1R1NOFORCE-DFMEA-01md	md	generated per deliverable-dfmea spec (qualitative version)
dfmea.pdf	VBF-T1R1NOFORCE-DFMEA-01pdf	pdf	md -> pdf exported by reportlab one-off script
delivery_index.md	VBF-T1R1NOFORCE-IDX-01md	md	generated per deliverable-package spec
delivery_index.pdf	VBF-T1R1NOFORCE-IDX-01pdf	pdf	md -> pdf exported by reportlab one-off script

Design summary

Endorsed design ****C7**** (round 6): Chen2020 NMC811/graphite baseline + single-crystal cathode 1.2 μm, fine graphite anode 0.8 μm (95 μm thick), high-transport electrolyte (sigma 1.8 S/m, D 5e-10 m²/s, t+ 0.5), pouch double-face liquid cold-plate cooling (h 25 W/m²K, A 0.2054 m²). All criteria pass: ED 457.93 Wh/kg; 4C@45 °C T_max 318.79 K, no plating (anode min +0.0082 V SPMe / +0.0052 V DFN); overcharge 4.7 V without thermal runaway (triggered=false). Audit chain: log.jsonl entries 0–25 (plan/funnel/propose/evaluate R1–R7/endorse/final); report.html rendered from log.